Interpreting Charts and Graphs

Bar Graphs and Pie Charts: Samples, Worked Examples and Practice Questions

English Communication Skills | B.Tech | Data and Reading Skills

Name:		Date:	
Class/Section:		Marks:	

Note: All data in this worksheet is illustrative and created for practice. Questions are labelled by skill: *Reading, Comparison, Calculation, Inference, Application* and *Critical thinking*. Answers are in the Answer Key at the end.

Part 1: How to Read a Chart

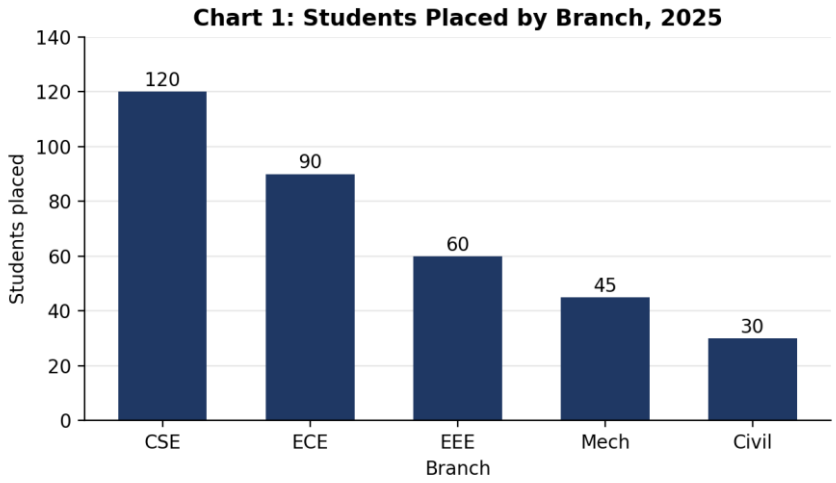
Before answering any question, follow these five steps:

Step	What to do
1. Title	Read the title. It tells you what the chart is about, where and when.
2. Labels and units	Check both axes (bar graph) or the labels (pie chart). Note the units: students, ₹, %, thousands.
3. Key or legend	If there is more than one colour or series, see what each stands for.
4. Highest, lowest, pattern	Find the biggest and smallest values and any trend, gap or change.
5. Fact or inference?	State what the chart shows (fact). Only then suggest reasons (inference) and use words like "may" or "possibly".

	Bar graph	Pie chart
Best for	Comparing values across categories or years	Showing parts of a whole
How to read	Read the height of each bar against the axis	Read the size of each sector as a % of the total
Key calculation	Difference, total, percentage change	Amount = % × total; angle = % × 360°
Watch out for	Axis that does not start at zero	Percentages must add up to 100%

Part 2: Bar Graphs

Sample Chart 1: Single Bar Graph



Branch	CSE	ECE	EEE	Mech	Civil
Students placed	120	90	60	45	30

Worked Examples (Chart 1)

Example A (Reading). Which branch had the highest placements? **Answer: CSE, 120.** Method: find the tallest bar and read its value.

Example B (Comparison). How many more students were placed from CSE than from Mech? **Answer: 75.** Method: subtract 120 − 45.

Example C (Calculation). What fraction of the total is ECE? **Answer: 90/345, about 26%.** Method: divide the value by the total (345) and multiply by 100.

Practice Questions on Chart 1

1.1. Which branch had the highest number of placements, and how many? *[Reading]*

1.2. Which branch had the lowest number of placements? *[Reading]*

1.3. How many more students were placed from ECE than from EEE? *[Comparison]*

1.4. Which branch had exactly half as many placements as CSE? *[Comparison]*

1.5. What was the total number of students placed in 2025? *[Calculation]*

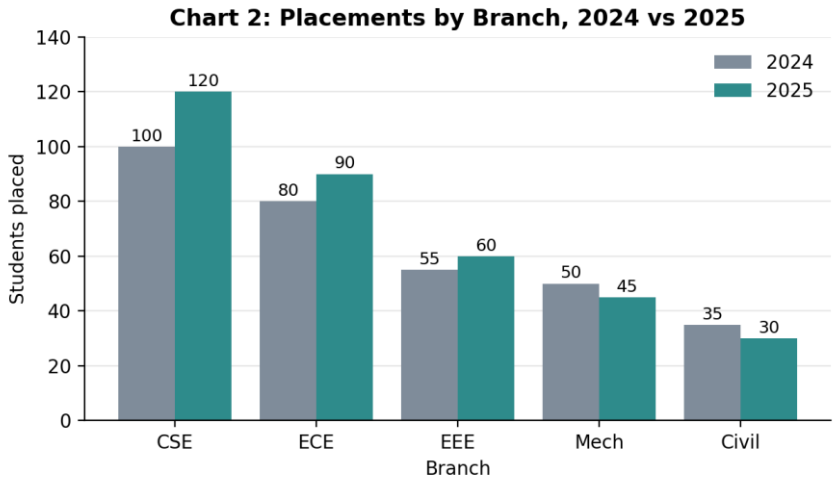
1.6. What percentage of all placed students came from CSE? (Answer to one decimal place.) *[Calculation]*

1.7. Did Mech and Civil together have more or fewer placements than ECE alone? By how many? *[Comparison]*

1.8. Suggest one possible reason why CSE had the most placements. *[Inference]*

Sample Chart 2: Grouped Bar Graph (Two Years)

A grouped bar graph compares two or more series side by side. Use the legend to see which colour is which year.



Branch	CSE	ECE	EEE	Mech	Civil	Total
2024	100	80	55	50	35	320
2025	120	90	60	45	30	345

> **Worked Example (Chart 2)**
> **Percentage change.** Formula: (new − old) ÷ old × 100.
> **Example:** ECE rose from 80 to 90. (90 − 80) ÷ 80 × 100 = **12.5% increase.**

Practice Questions on Chart 2

2.1. Which branch showed the largest increase in placements from 2024 to 2025? *[Reading]*

2.2. Which branches had fewer placements in 2025 than in 2024? By how many? *[Comparison]*

2.3. Find the total placements in each year. *[Calculation]*

2.4. What was the overall percentage increase in placements from 2024 to 2025? *[Calculation]*

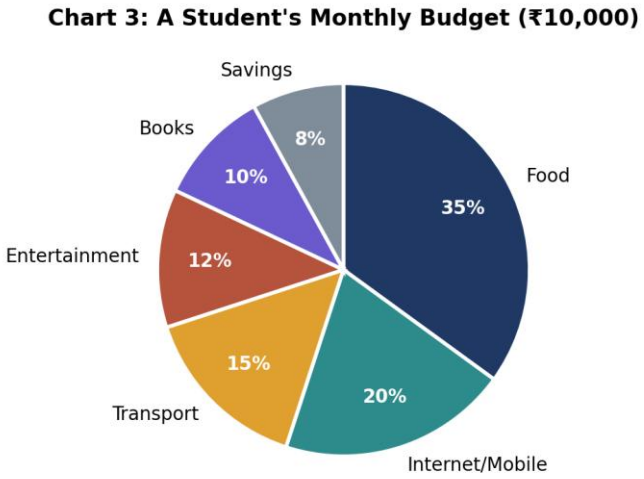
2.5. Calculate the percentage increase for CSE. *[Calculation]*

2.6. Which grew faster in percentage terms: ECE or EEE? Show your working. *[Calculation]*

2.7. A classmate says, "Civil students are less capable, because their placements fell." Is this conclusion supported by the graph? Explain. *[Critical thinking]*

Part 3: Pie Charts

Sample Chart 3: A Student's Monthly Budget



Item	Food	Internet/Mobile	Transport	Entertainment	Books	Savings
Share	35%	20%	15%	12%	10%	8%

Worked Examples (Chart 3)

Example A (Amount). How much is spent on Food? **Answer: 35% of ₹10,000 = ₹3,500.**

Example B (Angle). What angle does the Food sector make? **Answer: 35% × 360° = 126°.** (Shortcut: 1% = 3.6°.)

Example C (Check). 35 + 20 + 15 + 12 + 10 + 8 = **100%.** Always check that the parts add up to the whole.

Practice Questions on Chart 3

3.1. Which item takes the largest share of the budget? *[Reading]*

3.2. How much money is spent on Internet/Mobile? *[Calculation]*

3.3. What share of the budget, in per cent and in rupees, goes to Food and Transport together? *[Calculation]*

3.4. What angle will the "Savings" sector make at the centre of the pie chart? *[Calculation]*

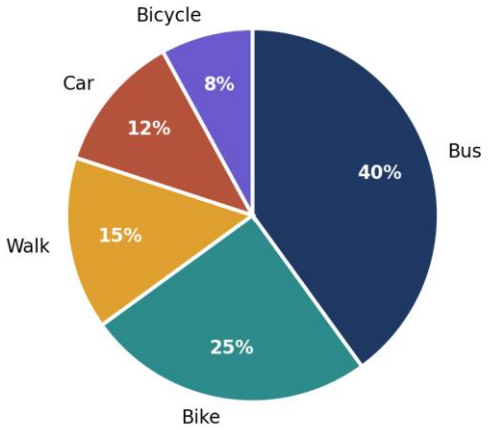
3.5. How much is spent on Entertainment and Books together? *[Calculation]*

3.6. Which items take up more than double the share of Savings? *[Comparison]*

3.7. The student wants to save ₹1,500 a month without changing the total budget. By how many percentage points must savings rise, and what could be reduced? *[Application]*

Sample Chart 4: How Students Travel to College

Chart 4: How 200 Students Travel to College



Mode	Bus	Bike	Walk	Car	Bicycle
Share	40%	25%	15%	12%	8%
Students (of 200)	80	50	30	24	16

Practice Questions on Chart 4

4.1. How many of the 200 students travel by bus? *[Calculation]*

4.2. How many more students come by bike than on foot? *[Comparison]*

4.3. What percentage of students use a motor vehicle (bus, bike or car)? *[Calculation]*

4.4. What angle will the "Car" sector make at the centre? *[Calculation]*

4.5. Which mode of travel is used by fewer than 20 students? *[Calculation]*

4.6. What can the pie chart NOT tell us about how students travel? Give two examples. *[Critical thinking]*

Part 4: Quick Check (Multiple Choice)

M1. Which chart is best for comparing values across different categories?
A) Pie chart B) Bar graph C) Neither D) Both equally

M2. In a pie chart, all the sectors together represent:
A) 50% B) 90° C) 100% (360°) D) 200%

M3. A sector with a central angle of 90° represents what percentage?
A) 10% B) 25% C) 40% D) 90%

M4. Which of these is an inference and not a fact from a chart?
A) 40% of students take the bus. B) The bus is the most used mode. C) Students prefer buses because they are cheaper. D) 8% of students cycle.

M5. The vertical axis of a bar graph starts at 80 instead of 0. What is the likely effect?
A) Differences look smaller B) Differences look exaggerated C) No effect D) Bars disappear

M6. Data from 200 students is shown in a pie chart. Walking is 15%. How many students walk?
A) 15 B) 20 C) 30 D) 45

Part 5: Writing Tasks

Task A. Write a summary of Chart 2 in about 80 words. Include an opening sentence, the main trend, one comparison and a concluding sentence.

Task B. Write a four-sentence description of Chart 4 for a college newsletter.

Useful Language

Purpose	Expressions
Introducing	The bar graph/pie chart shows, illustrates, compares, gives information about...
Highest/lowest	the highest figure, the largest share, the lowest number, the smallest proportion
Trends	rose sharply, climbed steadily, increased slightly, dipped, remained stable
Comparing	twice as much as, slightly higher than, far fewer than, the same as, in contrast
Proportion	accounts for, makes up, a quarter of, the majority of, just over half
Concluding	Overall, it is clear that... / In summary, the chart suggests that...

Answer Key

Chart 1: Single Bar Graph

1.1. CSE, with 120 students.

1.2. Civil, with 30 students.

1.3. 30 more (90 − 60 = 30).

1.4. EEE (60 is half of 120).

1.5. 120 + 90 + 60 + 45 + 30 = 345 students.

1.6. 120 ÷ 345 × 100 = 34.8%.

1.7. Fewer. 45 + 30 = 75, which is 15 fewer than ECE (90).

1.8. Sample answer: there may be higher demand for software professionals among recruiters. The graph shows numbers only, not reasons, so this is an inference, not a fact.

Chart 2: Grouped Bar Graph

2.1. CSE, an increase of 20 (100 to 120).

2.2. Mech (50 to 45) and Civil (35 to 30), each down by 5.

2.3. 2024: 100 + 80 + 55 + 50 + 35 = 320. 2025: 345.

2.4. (345 − 320) ÷ 320 × 100 = 25 ÷ 320 × 100 ≈ 7.8%.

2.5. 20 ÷ 100 × 100 = 20%.

2.6. ECE: 10 ÷ 80 × 100 = 12.5%. EEE: 5 ÷ 55 × 100 ≈ 9.1%. So ECE grew faster, even though both rose by a small number.

2.7. No. The graph shows only the number of placements, not the reasons. Placements could have fallen because fewer companies visited, fewer students were eligible, or market conditions changed. Ability cannot be concluded from this data.

Chart 3: Pie Chart (Budget)

3.1. Food (35%).

3.2. 20% of ₹10,000 = ₹2,000.

3.3. 35% + 15% = 50%, which is ₹5,000 (half the budget).

3.4. 8% of 360° = 28.8°.

3.5. 12% + 10% = 22% of ₹10,000 = ₹2,200.

3.6. Double the savings is 16%. Food (35%) and Internet/Mobile (20%) are both higher.

3.7. ₹1,500 is 15%, so savings must rise by 7 percentage points (8% to 15%), which is ₹700 more. Answers vary; for example, cut Entertainment from 12% to 5%, or reduce Internet/Mobile spending.

Chart 4: Pie Chart (Travel)

4.1. 40% of 200 = 80 students.

4.2. Bike: 25% = 50. Walk: 15% = 30. The difference is 20 students.

4.3. 40% + 25% + 12% = 77%.

4.4. 12% of 360° = 43.2°.

4.5. Bicycle: 8% of 200 = 16 students. (Car = 24 and Walk = 30 are both above 20.)

4.6. It cannot show how far students travel, how long the journey takes, what it costs, or why students choose each mode. It shows only the share of students using each mode.

Quick Check (MCQ)

M1. B) Bar graph

M2. C) 100%, or 360°

M3. B) 25% (90 ÷ 360 × 100)

M4. C). The chart does not state any reason.

M5. B) Differences look exaggerated, so always check where the axis begins.

M6. C) 30 (15% of 200)

Writing Tasks: Model Answers

Task A: Chart 2

The bar graph compares the number of B.Tech students placed in five branches in 2024 and 2025. Overall placements rose from 320 to 345, an increase of about 8 per cent. CSE showed the biggest gain, climbing from 100 to 120, while ECE and EEE also improved slightly. In contrast, Mechanical and Civil each fell by five students. Overall, placements grew in most branches, with computing leading the rise.

Task B: Chart 4

The pie chart shows how 200 students travel to college. The bus is the most popular mode, used by 40% of students, followed by bikes at 25%. Walking accounts for 15% and cars for 12%, while only 8% use a bicycle. Overall, more than three-quarters of students depend on motor vehicles.

Teacher's note: *Award marks for accuracy of figures, correct comparison language and a clear separation of fact from inference. Accept any reasonable inference in the Inference and Critical thinking questions if it is clearly labelled as a possibility.*